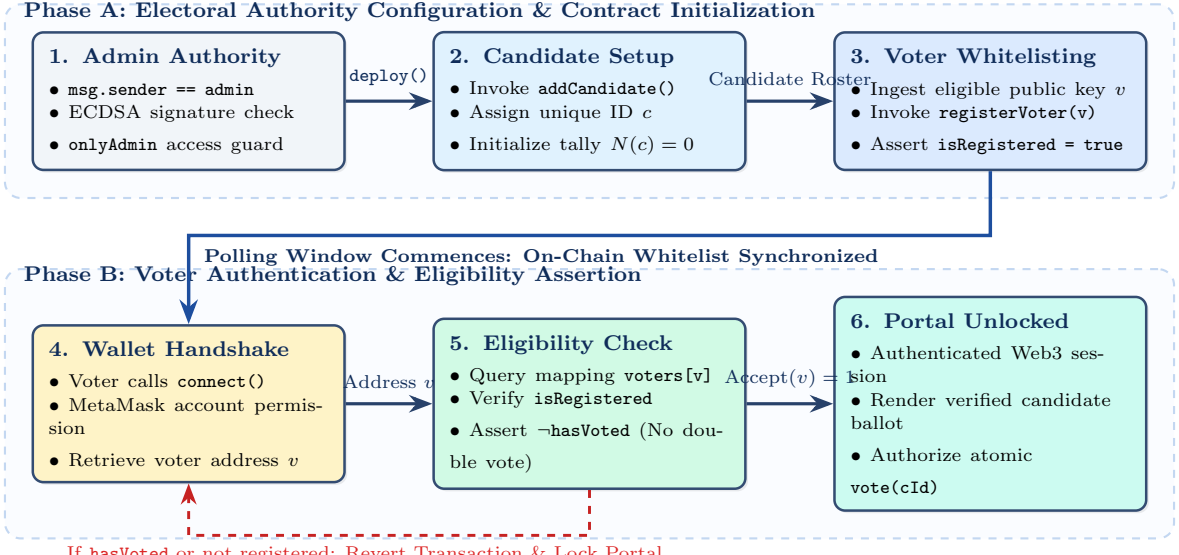

Phase A: Electoral Authority Configuration & Contract Initialization
1. Admin Authority
• msg.sender == admin
• ECDSA signature check
• onlyAdmin access guard
deploy()
2. Candidate Setup
• Invoke addCandidate()
• Assign unique ID c
• Initialize tally N(c) = 0
Candidate Roster
3. Voter Whitelisting
• Ingest eligible public key v
• Invoke registerVoter(v)
• Assert isRegistered = true
Polling Window Commences: On-Chain Whitelist Synchronized
Phase B: Voter Authentication & Eligibility Assertion
4. Wallet Handshake
• Voter calls connect()
• MetaMask account permission
• Retrieve voter address v
Address v
5. Eligibility Check
• Query mapping voters[v]
• Verify isRegistered
• Assert ¬hasVoted (No double vote)
Accept(v) = 1
6. Portal Unlocked
• Authenticated Web3 session
• Render verified candidate ballot
• Authorize atomic vote(cId)
If hasVoted or not registered: Revert Transaction & Lock Portal